Periodic cubic smoothing splines as a quadratic minimization problem

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Abstract

Smoothing cubic splines are an effective tool for approximating points in space by smooth curves. In this note we consider cubic splines with periodic conditions (closed curves) and we prove that the approximation problem can be written as a positive definite quadratic problem. Therefore, smoothing periodic cubic spline approximations can be conveniently obtained by solving a well-conditioned linear system.

Preliminaries: periodic cubic splines

Let us consider a closed connected interval $[a,b] \subset \mathbb{R}$ on the real line. A cubic spline f defined on the interval $[a,b]$ is a funciton $f \in C^2([a,b],\mathbb{R})$ that can be written as a 3rd-degree polynomial in a piecewise manner. The subintervals on which the spline f coincides with 3rd-degree polynomials are obtained by means of the so-called breaking sets.

Definition (Breaking set)**.** *Given a set o points $X = \{x_k\}_{k=1}^n \subset \mathbb{R}$, we say that X is a breaking set for $[a,b]$ if $x_k < x_{k+1}$ for every $k = 1,\dots,n-1$ and both $x_1 = a$ and $x_n = b$. A breaking set X consisting of n points determines $n-1$ interval lengths $h_k := x_{k+1} - x_k$, for $k = 1,\dots,n-1$.*

We are now able to give the following.

Definition (Cubic spline)**.** *Given $X = \{x_k\}_{k=1}^n$ a breaking set for $[a,b] \subset \mathbb{R}$, a function $f \in C^2([a,b],\mathbb{R})$ is a cubic spline with breaking set X if f can be written as a 3rd-degree polynomial on each interval $[x_k, x_{k+1}]$ with $k = 1,\dots,n-1$. In this case we write $f \in SP_X^3$.*

By definition, a cubic spline $f \in SP_X^3$ can be written as a 3rd-degree polynomial f_k over each interval $[x_k, x_{k+1}]$. Therefore, there exist $a_k, b_k, c_k, d_k \in \mathbb{R}$ such that

$$f(x) = f_k(x) = a_k + b_k(x - x_k) + c_k(x - x_k)^2 + d_k(x - x_k)^3 \qquad (1)$$

for $x \in [x_k, x_{k+1}]$ with $k = 1,\dots,n-1$. However, not for every choice of the coefficients $a_k, b_k, c_k, d_k \in \mathbb{R}$ a function f defined as in (1) is a cubic spline: the continuity of derivatives $f^{(0)}, f^{(1)}, f^{(2)}$ on $[a,b]$ imposes some constraints on the coefficients. These constraints can be expressed in $n-2$ equations each one corresponding to an interior break x_k, with $k = 2,\dots,n-1$. More specifically, in order to ensure continuity, equalities $f_k^{(j)}(x_{k+1}) = f_{k+1}^{(j)}(x_{k+1})$ must be verified for all $j = 0,1,2$ and $k = 1,\dots,n-2$. As we can see, no conditions need to be

imposed on interval ends x_1, x_n in order to get a valid cubic spline. Nevertheless, by extending continuity constraints to interval ends x_1, x_n, we obtain the so-called *periodic* cubic splines, i.e., cubic splines whose derivatives till order 2 coincide on interval ends.

By taking into account these considerations, we can give the following.

Definition (Periodic cubic spline)**.** *Given $X = \{x_k\}_{k=1}^n$ a breaking set for $[a,b] \subset \mathbb{R}$, a function $f \in C^2([a,b],\mathbb{R})$ is a periodic cubic spline with breaking set X if there exist $a_k, b_k, c_k, d_k \in \mathbb{R}$, with $k = 1, \ldots, n-1$, such that*

$$f(x) = f_k(x) = a_k + b_k(x - x_k) + c_k(x - x_k)^2 + d_k(x - x_k)^3 \tag{2}$$

for $x \in [x_k, x_{k+1}]$, and

$$f_k^{(j)}(x_{k+1}) = f_{k+1}^{(j)}(x_{k+1}) \qquad \text{for } j = 0, 1, 2, \tag{3}$$

with the convention that $f_n^{(j)}(x_n) = f_1^{(j)}(x_1)$. In this case we write $f \in SP_X^{3,per}$.

Continuity/periodic onditions in (3) can be expressed as a set of $3(n-1)$ equations by using polynomial expressions in (2) and performing derivatives. By doing so, we get

$$\begin{cases} a_{k+1} = a_k + b_k h_k + c_k {h_k}^2 + d_k {h_k}^3 \\ b_{k+1} = b_k + 2c_k h_k + 3d_k {h_k}^2 \\ c_{k+1} = c_k + 3d_k h_k \end{cases} \tag{4}$$

for $k = 1, \ldots, n-1$, with the convention that $a_n = a_1$, $b_n = b_1$ and $c_n = c_1$. By following the trick in [1], these constraints can be expressed in a convenient matricial form. Indeed, let us first rearrange the equations as

$$\begin{cases} b_k = \frac{a_{k+1} - a_k}{h_k} - c_k h_k - d_k {h_k}^2 \\ 2c_k h_k = b_{k+1} - b_k - 3d_k {h_k}^2 \\ d_k = \frac{c_{k+1} - c_k}{3h_k} \end{cases} . \tag{5}$$

Then, by plugging expressions for b_k, b_{k+1}, d_k in the second equation we find that coefficients a_k, c_k must verify

$$c_k h_k + 2c_{k+1}(h_k + h_{k+1}) + c_{k+2} h_{k+2} = 3\left[a_k \frac{1}{h_k} - a_{k+1}\left(\frac{1}{h_k} + \frac{1}{h_{k+1}}\right) + a_{k+2}\frac{1}{h_{k+1}}\right]. \tag{6}$$

for all $k = 1, \ldots, n-1$, where we recall the convention that $a_n = a_1$, $b_n = b_1$ and $c_n = c_1$. Thus, by denoting $a = (a_1, \ldots, a_{n-1})^T, c = (c_1, \ldots, c_{n-1})^T \in \mathbb{R}^{n-1}$, equations (6) can be arranged in a linear system involving a, c as

$$Sc = 3Va, \tag{7}$$

where the $(n-1) \times (n-1)$ matrices S and V are defined by

$$S = \begin{pmatrix} 2(h_{n-1} + h_1) & h_1 & & & & & h_{n-1} \\ h_1 & 2(h_1 + h_2) & h_2 & & & & \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ & & & h_{n-3} & 2(h_{n-3} + h_{n-2}) & h_{n-2} & \\ h_{n-1} & & & & h_{n-2} & 2(h_{n-2} + h_{n-1}) & \end{pmatrix}$$

$$V = \begin{pmatrix} -\frac{1}{h_{n-1}} - \frac{1}{h_1} & \frac{1}{h_1} & & & & & & \frac{1}{h_{n-1}} \\ \frac{1}{h_1} & -\frac{1}{h_1} - \frac{1}{h_2} & \frac{1}{h_2} & & & & & \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ & & & & \frac{1}{h_{n-3}} & -\frac{1}{h_{n-3}} - \frac{1}{h_{n-2}} & \frac{1}{h_{n-2}} & \\ \frac{1}{h_{n-1}} & & & & & \frac{1}{h_{n-2}} & -\frac{1}{h_{n-2}} - \frac{1}{h_{n-1}} & \end{pmatrix}$$

Indeed, for each $k = 1, \ldots, n-2$, the $(k+1)$-th row of the system is defined by equation (6). The 1st row is defined for $k = n-1$. Matrix V is symmetric, diagonally dominant with negative diagonal entries, thus negative semidefinite. Matrix S is symmetric, strictly diagonally dominant with positive diagonal entries, thus positive definite (and invertible). Therefore, given a set of coefficients $a \in \mathbb{R}^{n-1}$, we can find $c \in \mathbb{R}^{n-1}$ that verifies the constraints as $c = 3S^{-1}Va$. Moreover, from equations for b_k and d_k in (5) we can get also matrix expressions for the calculation of $b, d \in \mathbb{R}^{n-1}$ that verfy the constraints. Let us define $H := \mathrm{diag}(h_1, \ldots, h_{n-1})$ and

$$D := \begin{pmatrix} -1 & 1 & & & & \\ & -1 & 1 & & & \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ & & & & -1 & 1 \\ 1 & & & & & -1 \end{pmatrix},$$

then we easily have that $d = 1/3H^{-1}Dc$ and $b = H^{-1}Da - Hc - H^2d$. The discussion above provides a proof for the following.

Lemma. *Let us consider $X = \{x_k\}_{k=1}^n$ a breaking set for $[a, b] \subset \mathbb{R}$. Then, every $a \in \mathbb{R}^{n-1}$ determines (not necessarily in a unique way) a periodic cubic spline $f \in SP_X^{3,per}$ defined as*

$$f(x) = f_k(x) = a_k + b_k(x - x_k) + c_k(x - x_k)^2 + d_k(x - x_k)^3,$$

for $x \in [x_k, x_{k+1}]$ and $k = 1, \ldots, n-1$. The coefficients $c, d, b \in \mathbb{R}^{n-1}$ satisfying the continuity/periodic constraints are obtained as

- $c = 3S^{-1}Va$
- $d = 1/3H^{-1}Dc$
- $b = H^{-1}Da - Hc - H^2d$

and they depend only on a. To emphasize this fact, we also write $f = f_a$.

An approximation problem: smoothing periodic cubic splines

Assume we are given a set of observations $\{(x_k, y_k)\}_{k=1}^n \subset \mathbb{R}^2$, with $X = \{x_k\}_{k=1}^n$ a breaking set for $[a, b] \subset \mathbb{R}$ and $y_n = y_1$. We aim at finding a periodic cubic spline that minimizes the squared (weighted) distances between its values on the breaks and the observations. We impose smoothness by penalizing the squared 2nd-order derivative of the function over the interval $[a, b]$.

Problem (Smoothing periodic cubic splines)**.** *Given a set of two-dimensinal observations* $\{(x_k, y_k)\}_{k=1}^n \subset \mathbb{R}^2$ *with* $X = \{x_k\}_{k=1}^n$ *a breaking set for* $[a, b] \subset \mathbb{R}$ *and* $y_n = y_1$*, find* $f \in SP_X^{3,per}$ *that minimizes the funtional*

$$F(f) = p \sum_{k=1}^{n-1} \left[\frac{f(x_k) - y_k}{w_k} \right]^2 + (1-p) \int_a^b \left| f^{(2)}(x) \right|^2 dx, \tag{8}$$

where $p \in [0, 1]$ *is a smoothing parameter and* $w_1, \ldots, w_{n-1}$ *are weights, i.e.,*

- $w_k \in [0, 1]$ *for* $k = 1, \ldots, n-1$*,*
- $\sum_{k=1}^{n-1} w_k = 1$*.*

An $f = f_{a^*} \in SP_X^{3,per}$ *minimizing* $F(f)$ *is called a smoothing periodic cubic spline approximating the values* $Y = \{y_k\}_{k=1}^n$ *over the set of breaks* X*.*

Given the previous Lemma, the problem of finding a smoothing periodic cubic spline can be turned into a quadratic minimization problem. For convenience, let us call

$$L(f) := \sum_{k=1}^{n-1} \left[\frac{f(x_k) - y_k}{w_k} \right]^2 \qquad G(f) := \int_a^b \left| f^{(2)}(x) \right|^2 dx$$

so that the objective functional (8) can be written as $F(f) = pL(f) + (1-p)G(f)$. By definition of cubic spline $f(x_k) = f_k(x_k) = a_k$, thus $L(f)$ is better written as

$$L(f) = \sum_{k=1}^{n-1} \left[\frac{a_k - y_k}{w_k} \right]^2 = (a - y)^T W^{-2} (a - y)$$

where $y = (y_1, \ldots, y_{n-1})^T$ and $W = \mathrm{diag}(w_1, \ldots, w_{n-1})$. Moreover, using some of the properties given in the previous section, after some manipulations one can see that

$$G(f) = \sum_{k=1}^{n-1} \int_{x_k}^{x_{k+1}} \left| f_k^{(2)}(x) \right|^2 dx = \sum_{k=1}^{n-1} \frac{4}{3} h_k ({c_k}^2 + c_k c_{k+1} + c_{k+1}) = \frac{2}{3} c^T S c.$$

Recalling that $c = 3S^{-1}Va$, we finally get $G(f) = 6a^T V^T S^{-T} V a$. Hence, the objective functional is quadratic with respect to the a variable. Therefore, we have $F(f) = F(f_a)$ with

$$\begin{aligned} F(f_a) &= p(a - y)^T W^{-2} (a - y) + (1-p) 6 a^T V^T S^{-T} V a \\ &= \frac{1}{2} a^T U a - v^T a + r \end{aligned} \tag{9}$$

where

$$\begin{aligned} U &= 2pW^{-2} + 12(1-p) V^T S^{-T} V, \\ v &= 2pW^{-2} y \\ r &= p y^T W^{-2} y \end{aligned} \quad . \tag{10}$$

The matrix W^{-2} is obviously positive definite, whereas $V^T S^{-T} V$ is positive semidefinite, thus U is positive definite. As a consequence, the minimization

problem for obtaining the smoothing periodic cubic spline f_{a^*} approximating the values $Y = \{y_k\}_{k=1}^n$ over the set of breaks X is well posed and has a unique solution. We proved the following.

Lemma. *The smoothing periodic cubic spline f approximating the values $Y = \{y_k\}_{k=1}^n$ over the set of breaks X is given as $f = f_{a^*}$ where*

$$a^* = \underset{a \in \mathbb{R}^{n-1}}{\operatorname{argmin}} \frac{1}{2} a^T U a - v^T a.$$

Equivalently, $a^* \in \mathbb{R}^{n-1}$ is the unique solution of the linear system $Ua = v$.

Numerical implementation in Matlab®

As we can see, some matrix inversions are needed to solve the minimization problem. Theoretically, we need both to compute S^{-T} in order to define U, and also to solve the linear system $Ua = v$. However, in Matlab no explicit inversion needs to be done, as these tasks can be efficiently solved by means of left division operator. Indeed, the matrices involved are quite well structured and Matlab routines can handle them very efficiently. We just need to consider that:

- Being matrix S symmetric and positive definite, then also S^{-1} it is. Therefore, $S^{-T} = S^{-1}$ and in Matlab we can use $S \setminus V$ in place of $S^{-T}V$.
- Matrix U is symmetric and positive definite, thus the system $Ua = v$ can be efficiently solved using $a = U \setminus v$.

The interested reader can find an implementation in Matlab for the computation of the smoothing periodic cubic slines approximating data points in the N-dimensional space by following this url: `https://it.mathworks.com/matlabcentral/fileexchange/59463-smoothing-cubic-splines-with-periodic-conditions`

References

[1] Helmuth Späth and Jörg Meier. Flexible smoothing with periodic cubic splines and fitting with closed curves. *Computing*, 40(4):293–300, 1988.